Jaeyeon Kim

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Education

Carnegie Mellon University, Language Technologies Institute **Aug. 2025 – Present**

PhD Student, School of Computer Science *GPA:* 4.00/4.00

Advisor: Prof. Carlos Busso and Prof. Shinji Watanabe

Seoul National University, College of Liberal Studies **Mar. 2018 – Aug. 2024**

B.S. in Computer Science and Engineering and B.A. in Philosophy *GPA:* 4.11/4.30

2-year absence to fulfill military duty (Apr. 2022 *- Mar.* 2024*)*

Research Interest

I aim to build AI that perceives and reasons about the world through multiple modalities, bridging language and vision with auditory information. Specifically, I am interested in **(1) advancing audio understanding and reasoning in large audio and omni-modal language models** and **(2) improving the comprehension and generation of audio(-visual) scenes**, including spatial audio.

Research Experience

Audio and Acoustics Research Group, Microsoft Research **May 2026 – Aug. 2026**

Research Intern, Mentor: Dr. Hannes Gamper *Redmond, WA, USA*

- Worked on a unified framework for first-order ambisonics understanding, generation, and editing.

MSP Lab & WAV Lab, Carnegie Mellon University **Aug. 2025 – Present**

Graduate Research Assistant, Advisor: Prof. Carlos Busso and Prof. Shinji Watanabe *Pittsburgh, PA, USA*

- Developing post-training methods to improve auditory reasoning in large audio-language models.
- Developing self-supervised encoders for spatial audio representation learning.

Vision and Learning Lab, Seoul National University **Mar. 2024 – Jul. 2025**

Research Intern, Advisor: Prof. Gunhee Kim *Seoul, South Korea*

- Conducted research on video-to-spatial audio generation and on egocentric 3D visual span forecasting.
- Developed a benchmark for large audio-language models and co-organized the DCASE 2025 AudioQA Challenge based on this benchmark.

MAUM AI Inc. (Previously MindsLab Inc.) **Apr. 2022 – Jun. 2024**

AI Scientist (Apr. 2022 – Jun. 2023), Senior AI Scientist (Jul. 2023 – Jun. 2024), *Seongnam, South Korea*

- Alternative military service as industrial technical personnel (Apr. 2022 - Mar. 2024).
- Led development of Korean/English Conformer-Transducer ASR systems deployed in 4+ commercial services, achieving 40% relative CER reduction and 50%/75% lower latency/memory.
- Led audio team research projects on TTS, ASR, and other audio tasks.

Music and Audio Research Group, Seoul National University **Jul. 2021 – Mar. 2022**

Research Intern, Advisor: Prof. Kyogu Lee *Gwanggyo, South Korea*

- Explored self-supervised speech representations and spoken language understanding.

Publications (*Equal Contributions)

[1] **Jaeyeon Kim**, Sebastian Braun, Hannes Gamper. "Spatial Transfusion: From Structured Spatial Scene Understanding to Generation and Editing", *Under review at ICASSP 2027*

[2] **Jaeyeon Kim**, Chao-Han Huck Yang, Luoyi Zhang, Chan-Jan Hsu, Fernando Ruiloba Portilla, Jinchuan Tian, Shinji Watanabe, Carlos Busso. "Long Listening Thoughts: Eliciting Open Auditory Reasoning with Deliberative Perception and Cognitive Refinement", *in Findings of EMNLP*, 2026

[3] Chan-Jan Hsu, **Jaeyeon Kim**, Chao-Han Huck Yang, Shinji Watanabe, Hung-yi Lee, Carlos Busso. "Listen to the Latents: Self-Correcting Speech Recognition in Large Audio Language Models Through Hidden-State Interactions", *in Findings of EMNLP*, 2026

[4] Nahyun Lee, Dongkeun Yoon, Guijin Son, Geewook Kim, Dayoon Ko, Jeonghun Park, Haneul Yoo, Jaewon Cho, Junghun Park, Changyoon Lee, Kyochul Jang, **Jaeyeon Kim**, Eunsu Kim, Woojin Cho, Seungone Kim. "K-BrowseComp: A Web Browsing Agent Benchmark Grounded in Korean Contexts", *in Findings of EMNLP*, 2026

[5] Abinay Reddy Naini*, **Jaeyeon Kim***, Chao-Han Huck Yang, Shinji Watanabe, Carlos Busso. "Comparative Reasoning: Making an Audio Language Model Better at Comparing Emotions", *in Interspeech (Oral)*, 2026

[6] Ziyang Ma, Ruiyang Xu, Yinghao Ma, Chao-Han Huck Yang, Bohan Li, **Jaeyeon Kim**, Jin Xu, Jinyu Li, Carlos Busso, Kai Yu, Eng Siong Chng, Xie Chen. "The Interspeech 2026 Audio Reasoning Challenge: Evaluating Reasoning Process Quality for Audio Reasoning Models and Agents", *in Interspeech*, 2026

[7] **Jaeyeon Kim**, Heeseung Yun, Tony Woo, Chao-Han Huck Yang, Gunhee Kim. "WoW-Bench: Evaluating Fine-Grained Acoustic Perception in Audio-Language Models via Marine Mammal Vocalizations", *in Findings of ACL*, 2026

[8] **Jaeyeon Kim***, Heeseung Yun*, Gunhee Kim. "Towards Scene-Aware Video-to-Spatial Audio Generation", *IJCV*, 2026

[9] Chao-Han Huck Yang, Sreyan Ghosh, Qing Wang, **Jaeyeon Kim**, Hengyi Hong, Sonal Kumar, Guirui Zhong, Zhifeng Kong, S Sakshi, Vaibhavi Lokegaonkar, Oriol Nieto, Ramani Duraiswami, Dinesh Manocha, Gunhee Kim, Jun Du, Rafael Valle, Bryan Catanzaro. "Multi-Domain Audio Question Answering Benchmark Toward Acoustic Content Reasoning", *in ICASSP*, 2026

[10] Heeseung Yun, Joonil Na, **Jaeyeon Kim**, Calvin Murdock, Gunhee Kim. "Gaze Beyond the Frame: Forecasting Egocentric 3D Visual Span", *in NeurIPS (Spotlight)*, 2025

[11] **Jaeyeon Kim**, Heeseung Yun, Gunhee Kim. "ViSAGe: Video-to-Spatial Audio Generation", *in ICLR*, 2025

[12] **Jaeyeon Kim**, Minjeong Jeon, Jaeyoon Jung, Sang Hoon Woo, Jinjoo Lee. "EnCLAP++: Analyzing the EnCLAP Framework for Optimizing Automated Audio Captioning Performance", *in DCASE 2024 Workshop*, 2024

[13] **Jaeyeon Kim**, Jaeyoon Jung, Sang Hoon Woo, Minjeong Jeon, Jinjoo Lee. "Expanding on EnCLAP with Auxiliary Retrieval Model for Automated Audio Captioning", *DCASE2024 Challenge Technical Report* ***(Ranked 2nd in Task 6 Automated Audio Captioning)***, 2024

[14] **Jaeyeon Kim**, Injune Hwang, Kyogu Lee. "Learning Semantic Information from Raw Audio Signal Using Both Contextual and Phonetic Representations", *in ICASSP (Oral)*, 2024

[15] **Jaeyeon Kim**, Jaeyoon Jung, Jinjoo Lee, Sang Hoon Woo. "EnCLAP: Combining Neural Audio Codec and Audio-Text Joint Embedding for Automated Audio Captioning", *in ICASSP*, 2024

[16] Junhyeok Lee, Wonbin Jung, Hyunjae Cho, **Jaeyeon Kim**, Jaehwan Kim. "PITS: Variational Pitch Inference Without Fundamental Frequency for End-to-End Pitch-Controllable TTS", *in ICML Workshop on SPIGM*, 2023

Academic Service

Reviewer

- ICASSP (2026, 2027), ICML (2026), SLT (2026)

Organizer

- DCASE 2025 Task 5 Audio Question Answering, Interspeech 2026 Audio Reasoning Challenge

Scholarships and Awards

Summa Cum Laude, College of Liberal Studies, Seoul National University **Aug. 2024**

Best Thesis Award, Department of Computer Science and Engineering, Seoul National University **Aug. 2024**

Scholarship of 100 Years of Humanities (4-year full scholarship), Korea Student Aid Foundation **2018 – 2022**

Talent Award of Korea (50 outstanding high-school students nationwide), Minister of Education **Dec. 2017**

Additional Information

Language. Korean-Native, English-Fluent

Philosophy. I am interested in engaging in philosophical discussions on AI and deriving insights for AI research from philosophical perspectives. My focus lies particularly within the domains of philosophy of mind, philosophy of language, and Kantian ethics. You can explore my writings on philosophical discussions [here] (in Korean, titles also in English).